

Bounds on Sunflower-free Families

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1 Introduction

A collection $A_1, \dots, A_k$ of k sets is said to form a *k-sunflower* if all pairwise intersections among them are the same, that is, if $A_i \cap A_j$ is the same for all distinct i and j . The terminology arises from the visual similarity to a sunflower when such sets are depicted as a Venn diagram, with the common elements clumped together at the center and the sets arranged in a circular pattern around this shared center like petals:

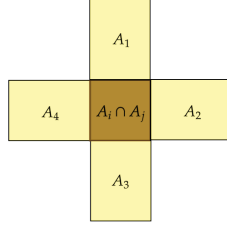


Figure 1: A visual depiction of a 4-sunflower as a Venn diagram

When $k = 3$, we simply say that the sets form a *sunflower*. A family of sets $\mathcal{F}$ is said to be *k-sunflower free* if no k of its members form a *k-sunflower*.

Introduced by Erdős and Rado, it is a longstanding problem in extremal combinatorics to bound the size of sunflower-free families. In this writeup, we'll be specifically interested in the case when $\mathcal{F}$ is a family of subsets of some fixed ambient ground set, which we may assume to be $\{1, \dots, n\}$. In this case, we have the trivial upper bound $|\mathcal{F}| \leq 2^n$, and it was conjectured by Erdős and Szemerédi in 1978 that the optimal upper bound is an exponential improvement upon this one:

Conjecture (Erdős and Szemerédi, 1978). *Let $\mathcal{F}$ be a k -sunflower-free family of subsets of $\{1, \dots, n\}$. Then*

$$|\mathcal{F}| \leq C^n$$

for some constant $C < 2$ depending only on k .

In full generality, this conjecture remains open even today. In recent years, however, the advent of powerful algebraic techniques has led to significant breakthroughs on the problem. In this writeup, we outline recent work by Naslund and Sawin [2] that establishes this conjecture asymptotically in the case $k = 3$. In particular, we employ a variant of the polynomial method to give a simple proof of the following:

Theorem 1 (Naslund and Sawin, 2017, [2]). *For any sunflower-free family $\mathcal{F}$ of subsets of $\{1, \dots, n\}$, we have that*

$$|\mathcal{F}| \leq 3(n+1) \sum_{k \leq \lfloor n/3 \rfloor} \binom{n}{k}.$$

In particular,

$$|\mathcal{F}| \leq \left(\frac{3}{2^{2/3}}\right)^{n(1+o(1))} \approx 1.89^{n(1+o(1))}.$$

2 The Slice Rank Polynomial Method

Our main tool for proving [Theorem 1](#) will be the *slice rank polynomial method*. If $\mathbb{F}$ is a field and $A \subset \mathbb{F}$, a function $f : A^k \rightarrow \mathbb{F}$ is said to be a *slice* if there exists $i \in \{1, \dots, k\}$ and functions $g : A \rightarrow \mathbb{F}$ and $h : A^{k-1} \rightarrow \mathbb{F}$ such that

$$f(x_1, \dots, x_k) = g(x_i)h(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_k).$$

In other words, f is a slice if it is possible to “slice out” and separate one of its inputs from the others. The *slice rank* of f is the smallest number r such that f can be written as a linear combination of r slices. The power of the notion of slice rank comes from the following:

Lemma (Slice Rank Lemma, [\[1\]](#)). *Let $\mathbb{F}$ be a field and $A \subset \mathbb{F}$ a finite subset thereof. If $T : A^k \rightarrow \mathbb{F}$ is a diagonal k -tensor with nonzero diagonal entries, then the slice rank of T is exactly $|A|$.*

Proof. We proceed by induction on k . When $k = 2$, the notion of slice rank is easily seen to coincide with the usual notion of rank, so in this case the slice rank of T is simply its number of diagonal entries, which by assumption is $|A|$. This proves the base case. Suppose now that $k > 2$ and that the claim has been proven for tensors of dimension $k - 1$. By writing

$$T(x_1, \dots, x_k) = \sum_{a \in A} \delta_a(x_1) T(a, x_2, \dots, x_k) \quad (1)$$

and observing that each term on the right-hand side is a slice, we see that the slice rank of T is at most $|A|$. To see that it is also at least $|A|$, suppose instead (for the sake of contradiction) that it is at most $|A| - 1$. Then it is possible to write

$$T(x_1, \dots, x_k) = \sum_{i=1}^k \sum_{\alpha \in I_i} f_{i,\alpha}(x_i) g_{i,\alpha}(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_k)$$

for some choice of sets $I_1, \dots, I_k$ of cardinalities adding up to at most $|A| - 1$, and some choice of functions $f_{i,\alpha} : A \rightarrow \mathbb{F}$ and $g_{i,\alpha} : A^{k-1} \rightarrow \mathbb{F}$. Consider the space V of functions $h : A \rightarrow \mathbb{F}$ that are orthogonal to all the $f_{k,\alpha}$ in the $L^2(A)$ -sense, that is,

$$V = \left\{ h : A \rightarrow \mathbb{F} : \sum_{x \in A} h(x) f_{k,\alpha}(x) = 0 \text{ for all } \alpha \in I_k \right\}.$$

Then V has dimension $d \geq |A| - |I_k|$. Moreover, observe that if we take a basis $\{h_1, \dots, h_d\}$ for V and arrange the h_i as the rows of a matrix, then we get a $d \times |A|$ matrix M of full rank d . Such a matrix must then contain d linearly independent columns $x_{i_1}, \dots, x_{i_d}$ which together give rise to an invertible $d \times d$ minor M' . By taking a general $h \in V$ and expanding it in terms of the basis $h = \lambda_1 h_1 + \dots + \lambda_d h_d$, we note that this minor M' has the effect of sampling h at the points $x_{i_1}, \dots, x_{i_d}$:

$$M' \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_d \end{bmatrix} = \begin{bmatrix} \lambda_1 h(x_{i_1}) + \dots + \lambda_d h(x_{i_1}) \\ \vdots \\ \lambda_1 h(x_{i_d}) + \dots + \lambda_d h(x_{i_d}) \end{bmatrix} = \begin{bmatrix} h(x_{i_1}) \\ \vdots \\ h(x_{i_d}) \end{bmatrix}.$$

In particular, because M' is invertible, we can solve the above equation for an $h \in V$ with arbitrary specified values at the $x_{i_1}, \dots, x_{i_d}$. By taking all the values to be nonzero, we can thus find a function $h \in V$ which is non-vanishing at $d \geq |A| - |I_k|$ points in A . If we now multiply both sides of [\(1\)](#) by $h(x_k)$ and sum over all x_k , then we find that

$$\sum_{x_k \in A} T(x_1, \dots, x_k) h(x_k) = \sum_{i=1}^{k-1} \sum_{\alpha \in I_i} f_{i,\alpha}(x_i) \widetilde{g_{i,\alpha}}(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_{k-1})$$

where $\widetilde{g_{i,\alpha}}(x_1, \dots, x_{k-1}) = \sum_{x_k \in A} g_{i,\alpha}(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_k) h(x_k)$. If we call the left-hand side $T'(x_1, \dots, x_{k-1})$, then we see that T' is a diagonal $(k-1)$ -tensor with at least $d \geq |A| - |I_k|$ nonzero diagonal entries. Hence by the inductive hypothesis its slice rank is at least $|A| - |I_k|$. On the other hand, by the expansion on the right-hand side, the slice rank of T' is at most $|A| - |I_k| - 1$. This contradiction completes the proof. $\square$

This lemma serves as the basis for the *slice rank polynomial method*, and plays an analogous role in the execution of this method to that played by Schwartz-Zippel in the ordinary polynomial method. The general strategy can be summarized as follows:

1. Take a set A that you are interested in bounding the size of;
2. Construct a suitable tensor $T : A^k \rightarrow \mathbb{F}$ (e.g. a multilinear polynomial) such that $T(x_1, \dots, x_k) \neq 0$ if and only if $x_1 = \dots = x_k$;
3. By the slice rank lemma, the slice rank of T is exactly $|A|$;
4. One can now bound $|A|$ by bounding the slice rank of T .

3 Proof of Theorem 1

We begin by identifying each $A \in \mathcal{F}$ with its indicator vector $v_A \in \{0, 1\}^n$, defined by having i th coordinate equal to one if $i \in A$ and zero otherwise. This gives us a collection of vectors $S \subset \{0, 1\}^n$ with $|S| = |\mathcal{F}|$, so it suffices to bound $|S|$ and work with these vectors.

With this reduction in place, for each $\ell = 1, \dots, n$ denote by S_ℓ the set of all members of S consisting of exactly ℓ ones. We claim that for any $u, v, w \in S_\ell$ that are not all equal, there exists i such that exactly two of u_i, v_i , and w_i are equal to one. To see this, note that this is certainly true in the case when the vectors are distinct, as this then merely amounts to the statement that the family of sets corresponding to S_ℓ is sunflower-free (which it is, because $\mathcal{F}$ is). The only case necessary to address, then, is when u, v , and w are non-distinct but not all equal, i.e. when exactly two of them are equal. We know, however, that each of the sets corresponding to a vector in S_ℓ has the same cardinality ℓ , so if without loss of generality $u = v \neq w$, then the sets corresponding to u and w cannot be subsets of one another; hence there must exist i such that $u_i = v_i = 1$ but $w_i = 0$. Thus the claim holds.

Consider, now, the multilinear polynomial $T : \{0, 1\}^n \times \{0, 1\}^n \times \{0, 1\}^n \cong \{0, 1\}^{3n} \rightarrow \mathbb{R}$ defined by

$$T(x, y, z) = \prod_{i=1}^n (2 - x_i - y_i - z_i).$$

By construction, T vanishes precisely on triples x, y, z for which there exists i such that exactly two of x_i, y_i , and z_i are equal to one. Equivalently, it is non-vanishing precisely on the triples that fail this property. By our work above, we know that for $x, y, z \in S_\ell$ the only such triples are trivial, i.e. $x = y = z$. Therefore, by applying the slice rank lemma to the restriction of T to $S_\ell \times S_\ell \times S_\ell$, we deduce that the slice rank of T is at least $|S_\ell|$. To obtain the result from here, we combine this estimate with a crude upper bound on the slice rank of T obtained by expanding T and grouping together terms in a clever way.

Specifically, notice that by expanding the product defining T we can write

$$T(x, y, z) = \sum_{I, J, K \subset [1, \dots, n], |I|+|J|+|K| \leq n} c_{I, J, K} x^I y^J z^K$$

where $x^I = \prod_{i \in I} x_i$ and similarly for y^J and z^K . Since $|I| + |J| + |K| \leq n$, at least one of $|I|, |J|$, or $|K|$ must be at most $n/3$. For each term in the above expansion, choose one of the monomials x^I, y^J , or z^K , making sure that whichever is chosen has total degree at most $n/3$. Finally, group together all terms with the same chosen monomial. By factoring out the common chosen monomial in each group, they become products of a function of a single variable with a function of the other variables, i.e. slices. The slice rank of T is therefore at most the number of groups; and since there is one group for each chosen monomial, we can bound the slice rank of T by the total number of possible choices. Indeed, for each of the three variables x, y and z , there are exactly $\sum_{k \leq \lfloor n/3 \rfloor} \binom{n}{k}$ choices that are theoretically possible, so

$$|S_\ell| \leq \text{slice rank of } T \leq 3 \sum_{k \leq \lfloor n/3 \rfloor} \binom{n}{k}.$$

It remains to observe that since $S = \bigcup_{\ell=0}^n S_\ell$, a union bound gives

$$|S| \leq 3(n+1) \sum_{k \leq \lfloor n/3 \rfloor} \binom{n}{k},$$

completing the proof of [Theorem 1](#).

The final remark follows by estimating the right-hand side using the binary entropy approximation

$$\binom{n}{\alpha n} = \Theta\left(\frac{2^{nH(\alpha)}}{\sqrt{n}}\right), \quad \text{where } H(\alpha) = -\alpha \log_2(\alpha) - (1 - \alpha) \log_2(1 - \alpha).$$

In our case, we have that

$$3(n+1) \sum_{k \leq \lfloor n/3 \rfloor} \binom{n}{k} \leq 3(n+1) \left(\frac{n}{3} + 1\right) \binom{n}{\lfloor n/3 \rfloor} \leq Cn^{3/2} 2^{nH(1/3)} = b^{O(\log_b n)} b^n = b^{n(1+o(1))}$$

where $b = H(1/3) = 3 \cdot 2^{-2/3}$.

References

- [1] Terence Tao. *A symmetric formulation of the Croot-Lev-Pach-Ellenberg-Gijswijt capset bound*. Blog post. May 2016. URL: <https://terrytao.wordpress.com/2016/05/18/a-symmetric-formulation-of-the-croot-lev-pach-ellenberg-gijswijt-capset-bound/> (visited on 03/20/2025).
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